

Urban Tree Crown Instance Segmentation Using Cross-Domain Foundation Models

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Abstract. Accurate urban individual tree crown segmentation remains challenging due to the complexity of urban environments as well as the unclear boundaries between adjacent trees and fused canopies. In this study, we propose a cross-domain two-step framework to accurately segment tree crowns in urban regions. 1) We apply a VLM-based approach for tree segmentation using GeoVLM. 2) We present cross-domain fine-tuning to Cellpose-SAM to adapt the segmentation model to tree structures. Experiments on the BAMFORESTS dataset outperform existing methods while requiring only limited fine-tuning. Experiments conducted in a selected urban environment further show that domain adaptation under complex condition works.

Keywords: Urban Trees · Cross-Domain · Individual Tree Crowns · Foundation Models

1 Introduction

Information at single tree level is relevant for urban micro climatic modeling, assessment of tree-level ecosystem services, as well as biodiversity analysis [9]. In urban regions, trees along streets have a direct impact on traffic safety [7]. This information is typically collected in cadastral data bases, whereas traditional data collection and regular updating is time-consuming and costly. Accurate instance segmentation of urban individual tree crowns (ITCs) from remote sensing (RS) imagery can help address this issue. However, it remains challenging due to the complex structure and similar visual characteristics of tree canopies [10]. Despite the rapid development in this field and the excellent ITC segmentation approaches proposed in recent years [1,5,10,16], existing methods typically rely on large-scale, manually annotated datasets. Furthermore, these learning-based approaches often struggle to generalize across domains, particularly when models trained on dense forest environments are applied to urban areas. Existing ITC benchmark datasets are primarily captured over forest and rural regions [10], whereas urban environments are considerably more complex with more deciduous trees and cluttered backgrounds, making urban ITC segmentation substantially more challenging. Training-free tree segmentation is proposed as an alternative solution to providing a model that eliminates the need for high quality annotation. Several approaches have been proposed for ITC segmentation [2,3].

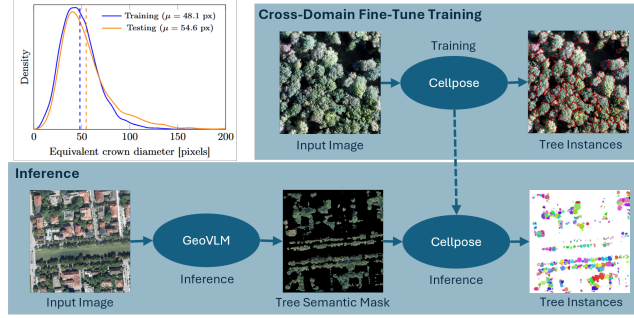


Fig. 1: Fine-tuning workflow and crown diameter distribution of BAMFORESTS. The dashed arrow indicates the fine-tuned CP-SAM model; the dashed line in the KDE plot indicates the mean crown diameter.

By combining state-of-the-art tree segmentation models trained on publicly available RS benchmark datasets with the cell and nucleus segmentation model Cellpose-SAM (CP-SAM) [12], preliminary results can be obtained. We systematically investigate and refine key parameters of CP-SAM that influence the ITC segmentation performance. To remove the influence of the complex background in urban region, we explore the potential of a zero-shot vision language model (VLMs) to achieve an improved urban tree semantic segmentation results.

2 Methodology

Figure 1 shows the cross-domain inference workflow using GeoVLM [4] and CP-SAM [12]. For urban scenes, GeoVLM first segments tree areas, followed by ITC delineation using the fine-tuned CP-SAM.

GeoVLM is a VLM for RGB RS imagery. It jointly processes imagery and natural-language prompts to perform tasks such as scene understanding, object localization, and semantic [4]. GeoVLM is employed to generate semantic tree masks, to solve the problem that CP-SAM is unable to separate trees in complex images with other objects. GeoVLM showed best performance compared to SOTA VLMs (SAM3, RemoteSAM and EarthMind) [17].

CP-SAM [12] combines the Segment Anything Model (SAM) with Cellpose, a biological foundation model originally developed for instance segmentation of biomedical cells [12]. Traditional segmentation models largely rely on finding distinct object edges or boundaries, however, CP-SAM reformulates the instance segmentation as a topological flow-convergence problem and mathematically forces the separation of touching objects even when no visual edge exists. ITCs and biological cells share similar geometric property which makes it high transferable to RS image based ITC instance segmentation.

Cellpose uses different parameters (see Table 1) to distinguish different cells and their sizes, which can be transferred to trees, their sizes and shapes. A key inference parameter is the diameter. According to literature, German tree crown diameters range from 4.39 m to 6.86 m in selected comparable studies [6,13,14].

Table 1: CP-SAM parameters used for training and inference output [11].

Parameter	Default	Training	Inference	Description
Diameter (px)	30	50	50	Approximate object diameter.
Cellprob thres.	0.0	0.0	-3.0	Minimum object probability for segmentation.
Flow threshold	0.4	0.4	1.0	Quality thres. for retaining masks.

The mean tree crown diameter of BAMFORESTS shown in Figure 1 aligns with these values. We calculate the corresponding CP-SAM diameter $D_{px} = \frac{D_c}{\text{GSD}}$, where D_c denotes the crown diameter in meters (m) and GSD is the ground sampling distance in meters per pixel (m/px).

3 Experiments and Results

CP-SAM was fine-tuned on the BAMFORESTS dataset [15]. The original imagery was down-sampled to 10 cm GSD, and the model was trained for 100 epochs with 100 out of 233 training data samples using the AdamW optimizer (learning rate 1×10^{-5} , weight decay 0.1) on image patches of size 512×512 pixels, following the official CP-SAM fine-tuning protocol. A crown diameter of 50 px was used as the default setting and evaluated together with lower and upper diameter values (see Table 2). The remaining parameters (Table 1) correspond to the train and inference settings and were experimentally selected using the validation data. Lower flow threshold values apply stricter quality control by rejecting more uncertain masks [11]. For testing, both Test-1 and Test-2 were combined [15].

The urban inference experiment focuses specifically on trees located along main streets with labels. The dataset captures a complex urban environment [8]. The RGB aerial images with 10 cm GSD were provided by the geodata service of the city of Munich including street trees, buildings, roads, vehicles, and other surrounding objects. Table 2(a) shows that CP-SAM can be successfully fine-tuned, with a diameter of 50 px yielding the best overall performance, which is consistent with the mean crown diameter shown in Figure 1. With an $F1$ of 83.07% the proposed model also outperforms the current BAMFORESTS baseline of 81% [5]. For inference evaluation, we consider one fully annotated tree section. As shown in Table 2(b), the fine-tuned model correctly segments 222 of 258 annotated crowns, substantially outperforming vanilla CP-SAM, which fails to produce reliable instance masks. The fine-tuned model yields 36 FN s (missed trees) and 92 FP s, mainly corresponding to vegetation or small spurious regions. Figure 2 partly shows the selected testing area. Figures (c–e) illustrate the inference workflow and the resulting ITCs. First, semantic segmentation is performed (c), followed by instance segmentation using the vanilla and fine-tuned CP-SAM models (d) and (e), respectively. Most false detections result from shadows and vegetation resembling tree crowns, as well as the selected CP-SAM inference settings, which allow the standard model in particular to generate

(a) BAMFORESTS test-set results (%).

Diameter	Model	IoU	F1	Precision	Recall
25 px	CP-SAM	4.65	7.89	68.30	4.78
	Fine-tuned	69.87	82.14	84.75	80.06
50 px	CP-SAM	9.33	14.72	59.33	9.87
	Fine-tuned	71.19	83.07	86.33	80.26
75 px	CP-SAM	19.25	28.32	66.70	21.38
	Fine-tuned	69.26	81.72	87.33	76.95

(b) Munich tree-crown instance evaluation using an IoU threshold of 0.5.

Metric	CP-SAM Fine-tuned	
Predicted instances	547	314
Matched instances (TP)	99	222
False positives (FP)	448	92
False negatives (FN)	159	36
Ground-truth instances	258	

Table 2: Quantitative comparison of vanilla and fine-tuned CP-SAM across crown-diameter settings on (a) BAMFORESTS and (b) Munich. Best results are shown in bold.



Fig. 2: (a) RGB image, (b) ground truth, (c) GeoVLM semantic segmentation, (d) ITC from vanilla CP-SAM, and (e) ITC from fine-tuned CP-SAM. (© Landeshauptstadt München, Kommunalreferat, GeodatenService 2021)

many small masks. These settings reflect a trade-off between detecting as many tree crowns as possible and limiting false detections.

4 Conclusion

In conclusion, the proposed cross-domain learning approach provides an effective solution for urban ITC segmentation while reducing the reliance on costly manual annotations. Zero-shot CP-SAM exhibits robust generalization ability, yet the lack of tree-specific knowledge leads to unstable performance, especially in the complex urban region. Despite using only a limited number of training samples, fine-tuned CP-SAM outperforms the current state-of-the-art baseline on BAMFORESTS. For urban imagery containing buildings, vehicles, and other non-tree objects, GeoVLM semantic masks suppress background clutter and enable CP-SAM to perform tree-specific instance segmentation on unseen data. These results provide practical guidance for adapting foundation models to data-limited urban tree-monitoring applications.

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